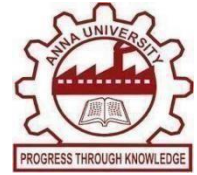




AN IoT BASED SYSTEM FOR MANAGING AND MONITORING SMART PLANT WATERING SYSTEM



A PROJECT REPORT

Submitted by

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In partial fulfillment for the award of the degree of

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ANNA UNIVERSITY : CHENNAI

BONAFIDE CETIFICATE

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EXTERNAL EXAMINER

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ABSTRACT

"An IoT, based system For Managing and Monitoring Smart Plant Watering System" is developed to create an automated irrigation mechanism which turns the pumping motor ON and OFF on detecting the moisture content of the earth using the soil moisture sensor without the intervention of human. The benefit of employee, these techniques is to decrease human interference and it is quite feasible and affordable. This smart irrigation system project is using devices that is programmed to collect the input signal according to moisture content of the soil and its output is given to the op-amp that will operate the pump.

The benefits of this IoT solution include significant water conservation, improved crop yield, reduced operational costs, and enhanced sustainability in agriculture. By ensuring precise and timely water delivery tailored to specific crop needs, the smart irrigation system contributes to environmental preservation and economic viability for farmers.

In conclusion, the IoT-based smart irrigation system represents a transformative advancement in agriculture, offering scalable solutions to address water scarcity challenges and promote efficient resource management in farming practices.

Keywords: *Soil moisture, Relay module, Arduino microcontroller.*

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CHAPTER 1

INTRODUCTION

The rapid advancement of technology in recent years has led to the development of the Internet of Things (IoT), which refers to the interconnection of various devices and systems through the internet, enabling them to collect and exchange data. This technological evolution has paved the way for innovative solutions across multiple domains, including agriculture and home gardening.

One such innovation is the IoT-based smart plant watering system, which aims to optimize plant care through automation and data-driven decision-making.

An IoT-based smart plant watering system integrates sensors, controllers, and communication devices to monitor and manage the watering needs of plants. By continuously collecting data on soil moisture levels, temperature, humidity, and other environmental factors, the system can determine the optimal times and amounts of water needed for different plants. This automated approach not only ensures that plants receive adequate water but also conserves resources by preventing overwatering and reducing water wastage.

The implementation of smart plant watering systems offers several benefits. For home gardeners, it provides a convenient and efficient way to maintain healthy plants, even when they are away from home. For larger-scale agricultural operations, it can lead to significant improvements in crop yield and resource management. Additionally, these systems contribute to sustainable gardening practices by optimizing water usage and minimizing the environmental impact of irrigation.

This report depicts the design and development of an IoT-based smart plant watering system. It explores the system architecture, including the hardware components such as sensors and microcontrollers, and the software aspects involving data processing and decision algorithms. Furthermore, the report discusses the challenges encountered during the development process and presents the results in various scenarios. Ultimately, this study aims to demonstrate the potential of IoT technology to enhance plant care and promote efficient water use in both domestic and agricultural settings.

1.1. SCOPE OF THE PROJECT

The scope of the project "An IoT-based system for managing and monitoring smart plant watering system" encompasses the development of a sensor-driven, automated and deployment irrigation system designed to optimize water usage for plant care. This system integrates soil moisture sensors, temperature and humidity sensors, and a central microcontroller connected to a cloud platform via the Internet of Things (IoT). The microcontroller will analyze real-time data from the sensors to determine the precise watering needs of the plants, triggering the irrigation system accordingly. The project also includes a user-friendly mobile application for remote monitoring and control, allowing users to set parameters, receive notifications, and view historical data on plant health and watering schedules. The ultimate aim is to create an efficient, sustainable, and user-friendly solution that conserves water while ensuring optimal plant growth.

1.2. OBJECTIVE OF THE PROJECT

Develop a with IoT technology to automate and optimize plant watering. The system aims to ensure plants receive adequate water based on real-time environmental conditions, thereby improving plant health and conserving water.

1.3. REVIEW SUMMARY

The project report is organized as follows: the chapter 2 narrates the literature survey, chapter 3 explains the existing system and methodologies, chapter 4 explains the proposed system and principles, chapter 5 depicts the architecture and design of the proposed methodology, chapter 6 provides the system and software requirements and installation methods, chapter 7 shows the proposed system implementation and the results of the proposed system, chapter 8 details the conclusion and future scope of the system and the last section provides the references and appendix of the proposed system.

CHAPTER 2

LITERATURE SURVEY

There are multiple other recent publications and surveys on IoT-based smart plant watering system using various other approaches are compiled.

1. Design and implementation of an IoT-based smart irrigation system

Year: 2019

Authors: N. Lakshmanaprabu, K. Shankar, R. Khanna, *et al.*

Summary:

Objective: To develop a smart irrigation system that minimizes water usage while ensuring optimal plant growth.

Methodology: Utilizes soil moisture sensors, temperature sensors, and a microcontroller to monitor environmental conditions. Data is transmitted via Wi-Fi to a cloud server for analysis.

Key Findings: The system can reduce water consumption by up to 30% compared to traditional irrigation methods.

Conclusion: IoT-based systems are effective in conserving water and ensuring efficient plant care.

2. IoT-based automated plant watering system using Raspberry Pi

Year: 2019

Authors: M. A. Mazidi, S. Naimi, *et al.*

Summary:

Objective: To automate plant watering using a Raspberry Pi to reduce human intervention.

Methodology: Integrates soil moisture sensors with a Raspberry Pi, which controls the water pump based on real-time data.

Key Findings: The system provides a reliable and cost-effective solution for home gardeners and small-scale farmers.

Conclusion: The automation significantly reduces the labor required for plant care while maintaining optimal soil moisture levels.

3. Smart irrigation system using IoT and wireless sensor networks

Year: 2020

Authors: S. K. Gupta, R. P. Sharma, *et al.*

Summary:

Objective: To develop a smart irrigation system leveraging IoT and wireless sensor networks to monitor and manage water distribution.

Methodology: Combines wireless sensors, a central control unit, and IoT connectivity to automate and optimize watering schedules.

Key Findings: The system improves water use efficiency and can be adapted for various types of crops.

Conclusion: Wireless sensor networks enhance the scalability and flexibility of IoT-based irrigation systems.

4. IoT-Enabled smart agriculture monitoring system using soil moisture sensors

Year: 2018

Authors: T. M. Nguyen, B. L. Lim, *et al.*

Summary:

Objective: To monitor soil moisture levels continuously and adjust watering schedules automatically.

Methodology: Deploys soil moisture sensors and an IoT gateway to collect and transmit data to a cloud platform for analysis.

Key Findings: Continuous monitoring helps in timely decision-making, reducing water wastage.

Conclusion: IoT-enabled monitoring systems provide precise control over irrigation, enhancing crop yield and water management.

5. An IoT-Based smart garden with automated watering system

Year: 2020

Authors: A. Verma, P. Singh, *et al.*

Summary:

Objective: To create an IoT-based smart garden that automates watering based on plant needs.

Methodology: Uses sensors to measure soil moisture and environmental conditions, with a microcontroller managing the irrigation system.

Key Findings: Automated systems ensure plants receive the right amount of water, promoting healthier growth.

Conclusion: Smart gardens can greatly benefit urban environments where water conservation is crucial.

CHAPTER 3

EXISTING SYSTEM

Currently, there are several commercially available automated irrigation systems that use various technologies and sensors to control irrigation. Most of these systems use a timer-based approach, which waters the plants at regular intervals, regardless of the soil moisture levels. This approach can lead to over-watering and waste of water resources.

Some existing systems use soil moisture sensors to determine the soil moisture levels and trigger irrigation when the levels fall below a certain threshold. However, these systems can be expensive and complex to install and maintain. Another technology used in automated irrigation systems is the use of weather data to optimize irrigation. This technology uses weather forecasts and historical data to predict the water requirements of plants and adjust irrigation accordingly. However, this technology requires sophisticated sensors and weather forecasting models, making it costly and complex to implement.

In our project, we are proposing an automated irrigation system that uses a rain sensor and a moisture sensor to control irrigation. The rain sensor is used to detect the presence of rain and suspend irrigation to prevent over-watering, while the moisture sensor is used to determine the soil moisture levels and trigger irrigation when the levels fall below a certain threshold. Our system offers a low-cost and practical solution that can be easily installed and maintained by homeowners and small-scale farmers.

CHAPTER 4

PROPOSED SYSTEM

Agriculture is a crucial sector of the economy in many countries. However, conventional irrigation systems are not efficient in terms of water usage and control. To overcome these issues, an IoT-based smart irrigation system is proposed in this project. The proposed system for automatic irrigation using IoT is an intelligent and efficient system that utilizes various sensors, microcontrollers, and Arduino Integrated Development Environment (IDE) technologies to monitor the moisture level of the soil, detect rainfall, and control the irrigation system. The proposed system for automatic irrigation using IoT not only overcomes the drawbacks of traditional irrigation systems, but it also incorporates additional features to improve water usage efficiency. One such feature is the use of a rain sensor to prevent over-watering and water wastage. The rain sensor detects the presence of rain and sends a signal to the microcontroller to stop the water pump, thereby preventing over-watering and water wastage. This feature ensures that water is only used when it is necessary and promotes efficient water usage.

Furthermore, the proposed system allows farmers to monitor their fields remotely through the Arduino (IDE) platform. The sensor data collected by the system is uploaded to the Arduino IDE, and farmers can access the data using their mobile devices. This allows farmers to monitor the soil moisture level, temperature, humidity, and rainfall in real-time, enabling them to make informed decisions regarding irrigation and crop management.

Overall, the proposed system for automatic irrigation using IoT is an intelligent and comprehensive solution that not only overcomes the drawbacks

of traditional irrigation systems but also incorporates additional features to promote efficient water usage and enable remote monitoring. With the help of this system, farmers can optimize their irrigation systems, reduce water wastage, and improve crop yield

COMPONENTS USED:

1. Arduino UNO board
2. Rain sensor
3. Humidity sensor
4. Relay Modules
5. Soil moisture sensor
6. power supply
7. Water pump

CHAPTER 5

MODEL

1. **Hardware:** This includes sensors to measure soil moisture, temperature. It also includes water pump or valves to control the watering system.
2. **Microcontroller:** A microcontroller like Arduino UNO can be used to interface with the sensors to collect data, and communicate with the internet.
3. **Internet Connectivity:** The system needs to be connected to the internet to send data to a cloud server and receive commands from a mobile app. This can be done using Wi-Fi, Ethernet, or other wireless technologies.
4. **Cloud Server:** Data collected from the sensor is sent to a cloud server for storage and analysis. The server also hosts the application programming interface (API) for communication with the mobile app.
5. **Mobile App:** The mobile app allows users to monitor the status of their plants, receive notifications when watering is needed, and control the watering system remotely.
6. **Integration:** The mobile app needs to be integrated with the cloud server's API to send and receive data.
7. **User Interface:** The mobile app should have a user-friendly interface that allows users to easily interact with the system and view data about their plants.
8. **Security:** Since the system is connected to the internet, security measures need to be implemented to protect against unauthorized access and ensure the privacy of user data.

5.1. MODEL DESCRIPTION

This prototype aims to saving time and avoiding problems like constant vigilance. It also helps in water conservation by automatically providing water to the plants or gardens depending on their water requirements. As technology is advancing, there is always a chance of reducing risks and making work simpler. Embedded and micro controller systems provide solutions for many problems.

This application precisely controls water system for gardens by using a sensor micro controller system. It is achieved by installing sensors in the field to monitor the soil temperature and soil moisture which transmits the data to the microcontrollerfor estimation of water demands of plants.

The soil moisture sensor senses the amount of moisture content in the soil which is uploaded to the Arduino board. The Arduino UNO board transfers the control over the system to the relay module which is responsible for switching operations. The relay module ensures proper irrigation of the field turning it on when the value of moisture is below the threshold value and turns off the supply when the moisture content is sufficient for the crop or plant thereby preventing under irrigation or over irrigation. The state of the relay module is indicated by the Light emitting diode (LED). A simple process flow representation of the system is shown in fig. 5.1. From the representation, it is clearly understood that the working of the system is simple and can be controlled easily.

Efficient and effective monitoring systems have an impact on the development and growth of plants and are highly vital in designing an effective irrigation control system in order to enhance the production of food with minimum water loss. Monitoring in the particular context of precision irrigation inculcates collecting data, which adequately leads to reflect the real-time status of the plant, soil, and weather of irrigation areas through the use of the Internet of Things (IoT) and Wireless Sensor Networks (WSN). In order to establish a real-time system of monitoring, IoT has led to establish a low-cost technology method that leads to improve the control and monitoring system for the irrigation process. Also, WSN also significantly contributes in real-time monitoring for precision farming. In this technique, a network of wireless sensor nodes is developed in order to sense, compute, and transmit information of different parameters.

5.1. MODEL ARCHITECTURE

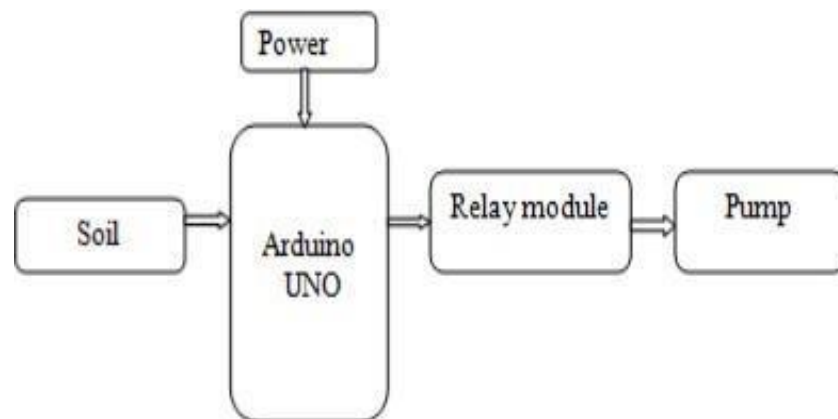


Fig. 5.1 Model Architecture

5.2.MODEL OPERATION

PROCEDURE:

1. Design and Planning:

- **Identify Requirements:** Determine the scale of irrigation needs (e.g., garden,farm, park).
- **Select Sensors:** Choose appropriate IoT sensors (e.g., soil moisture, weather,temperature).
- **Choose Communication Protocol:** Decide on the IoT communication protocol(e.g., Wi-Fi, LoRa, NB-IoT) based on range and data requirements.

2. Sensor Deployment:

- **Install Soil Moisture Sensors:** Place sensors in key locations to monitor soilmoisture levels accurately.
- **Weather Sensors:** Deploy weather sensors to gather real-time data ontemperature, humidity, and precipitation.
- **Connect Sensors:** Ensure all sensors are connected to a central IoT hub orgateway.

3. IoT Gateway Setup:

- **Select Gateway Device:** Choose a suitable IoT gateway (e.g., Raspberry Pi,Arduino with Wi-Fi shield).
- **Connect Sensors to Gateway:** Configure sensors to communicate data to thegateway.
- **Set Up Connectivity:** Establish connectivity with the IoT platform or cloud service for data storage and analysis.

4. Data Processing and Analysis:

- **Data Collection:** Collect and store sensor data in the cloud or a local server.
- **Data Analysis:** Implement algorithms to analyze soil moisture, weather forecasts, and historical data to optimize irrigation schedules.

5. Decision Making and Control:

- **Automation Rules:** Define rules and thresholds for automated irrigation decisions based on sensor data (e.g., activate irrigation if soil moisture drops below a certain level).
- **Control Mechanism:** Integrate actuators (e.g., solenoid valves) controlled by the IoT platform to turn irrigation systems on/off.

6. User Interface and Monitoring:

- **Dashboard Development:** Create a user-friendly interface (web or mobile app) to monitor system status and receive alerts.
- **Visualization:** Display real-time and historical data (e.g., moisture levels, irrigation events).

7. Implementation and Testing:

- **Install Irrigation Components:** Set up and test irrigation equipment connected to the IoT system.
- **Simulation:** Conduct simulations to verify system functionality and responsiveness to varying conditions.

8. Maintenance and Optimization:

- **Regular Monitoring:** Monitor system performance regularly to ensure sensors are functioning correctly and data is accurate.
- **Fine-tuning:** Adjust irrigation schedules and thresholds based on feedback and analysis to optimize water usage efficiency.

9. Security Considerations:

- **Data Security:** Implement security measures (e.g., encryption, authentication) to protect IoT devices and data from unauthorized access.
- **Update Firmware:** Keep firmware and software up to date to address security vulnerabilities.

10. Scaling and Expansion:

- **Future Expansion:** Plan for scalability to accommodate additional sensors or expand coverage area as needed.
- **Integration:** Consider integrating with other smart agriculture systems or IoT applications for comprehensive farm management.

CHAPTER 6

SYSTEM SPECIFICATION

6.1. Hardware Requirements

1. Arduino UNO board
2. Rainfall sensor
3. Humidity sensor
4. Relay Modules
5. Soil moisture sensor
6. Power Supply
7. Water pump

6.2. Software Requirements

1. Arduino IDE
2. Arduino Programming Language
3. Water pump Control Logic

Arduino IDE

The Arduino Integrated Development Environment (IDE) is the primary software tool used for programming the Arduino UNO. It provides a user-friendly interface to write, compile, and upload code to the Arduino board.

Timing Libraries

To manage the water pump control timings, use the built-in "millis()" function in Arduino. This function helps create precise timing intervals without blocking the program execution.

Sensor Libraries

Depending on the type of weather detection sensor (e.g., Humidity sensor, soil moisture sensors, Rain sensor etc.), utilize specific sensor libraries to interface and read data from those sensors.

6.3. Simulation Diagram

The simulation diagram for the proposed system is shown in fig.6.1.

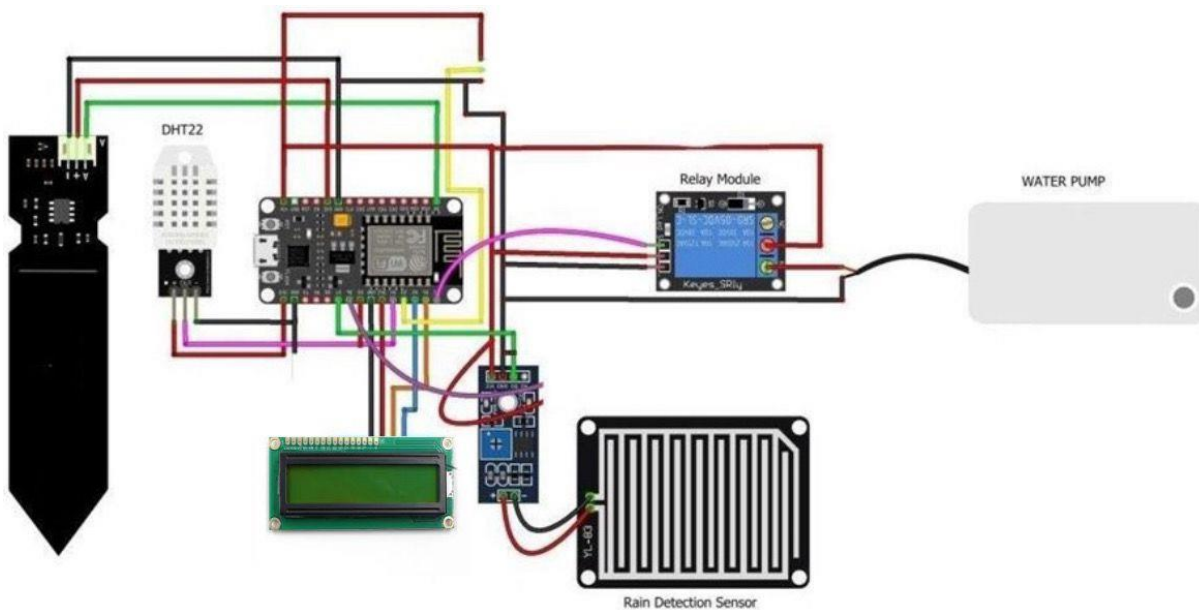


Fig. 6.1 Simulation Diagram

DHT11 Sensor: Data pin connected to digital pin 2 on the Arduino. Rain Sensor: Signal pin connected to digital pin 7 on the Arduino. Soil Moisture Sensor: Analog output pin connected to analog pin A0 on the Arduino. LCD Display: Register Select pin connected to digital pin 12 on the Arduino. Enable pin connected to digital pin 11 on the Arduino. D4 pin connected to digital pin 5 on the Arduino. D5 pin connected to digital pin 4 on the Arduino. D6 pin connected to digital pin 3 on the Arduino. D7 pin connected to digital pin 2 on the Arduino.

CHAPTER 7

IMPLEMENTATION AND RESULT

Implementing smart irrigation in agriculture involves the integration of advanced sensors and automated systems to optimize water usage. This process starts with assessing the specific needs of the crops and the environmental conditions, followed by installing soil moisture sensors, weather stations, and smart controllers.

These devices collect real-time data, which is then analyzed to make precise irrigation decisions, adjusting water delivery based on soil moisture levels and weather forecasts. The results of such implementation are significant: farmers typically see a reduction in water usage by up to 50%, leading to lower water bills and operational costs. Moreover, crops benefit from optimal watering, which can enhance growth, increase yields by up to 20%, and improve crop quality. The environmental impact is also positive, with reduced water waste and soil erosion, contributing to more sustainable farming practices. The testing model is shown in fig. 7.1.

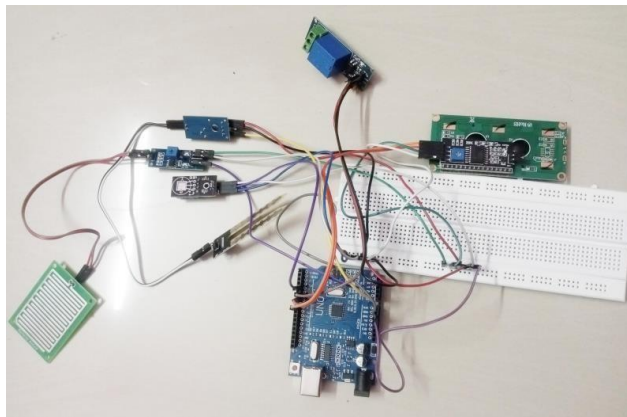


Fig. 7.1. Testing model

CHAPTER 8

CONCLUSION AND FUTURE SCOPE

CONCLUSION:

In conclusion, smart irrigation represents a significant advancement in agricultural water management, combining technology and data analytics to optimize water use. By implementing smart irrigation systems, farmers can achieve substantial water savings, reduced operational costs, and improved crop yields and quality. These systems provide precise and efficient irrigation tailored to the specific needs of the crops and prevailing weather conditions, leading to better resource conservation and environmental sustainability. As the agricultural sector continues to face challenges such as water scarcity and climate change, smart irrigation offers a practical and effective solution for enhancing productivity and ensuring the long-term viability of farming practices.

FUTURE SCOPE:

The future scope of smart irrigation is expansive and promising, driven by advancements in technology and increasing awareness of sustainable water management. Future developments may include the integration of artificial intelligence and machine learning to enhance predictive analytics, allowing for even more precise and proactive irrigation management. The use of drones and satellite imagery could further refine the monitoring of crop health and soil moisture levels across large agricultural areas, providing real-time data for more informed decision-making. Additionally, the adoption of IoT technology will continue to improve system connectivity and efficiency, enabling seamless communication between various sensors and controllers.

With the growing emphasis on sustainable agriculture, smart irrigation systems are likely to become more affordable and accessible, encouraging widespread adoption. Innovations in renewable energy sources, such as solar-powered irrigation systems, will also contribute to the eco-friendliness of these solutions.

Ultimately, the future of smart irrigation holds great potential for enhancing water conservation, increasing agricultural productivity, and promoting environmental sustainability on a global scale.

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APPENDIX – 1 CODING

```
#include
<LiquidCrystal.h>
#include <DHT.h>
#define DHTPIN 2
#define DHTTYPE DHT11
DHT dht(DHTPIN, DHTTYPE);
const int rainPin = 7;
const int soilPin =
A0;
LiquidCrystal lcd(12, 11, 5, 4, 3,
2);void setup() {
Serial.begin(9600);
lcd.begin(16, 2);
dht.begin();
}
void loop() {
int          soilValue          =
analogRead(soilPin);int rainValue
=    digitalRead(rainPin);    float
humidity = dht.readHumidity();
lcd.clear();
```

```

lcd.setCursor(0, 0);
lcd.print("Soil Moisture:
");lcd.print(soilValue);
lcd.setCursor(0, 1);
lcd.print("Rain: ");
lcd.print(rainValue);
lcd.setCursor(8, 1);
lcd.print("Humidity: ");
lcd.print(humidity);
if (soilValue < 500 && rainValue == LOW && humidity < 80) {
lcd.setCursor(0, 3);
lcd.print("Watering...");
  } else {
    lcd.setCursor(0,
3);
    lcd.print("Not Watering");
  }
  delay(1000);
}

```